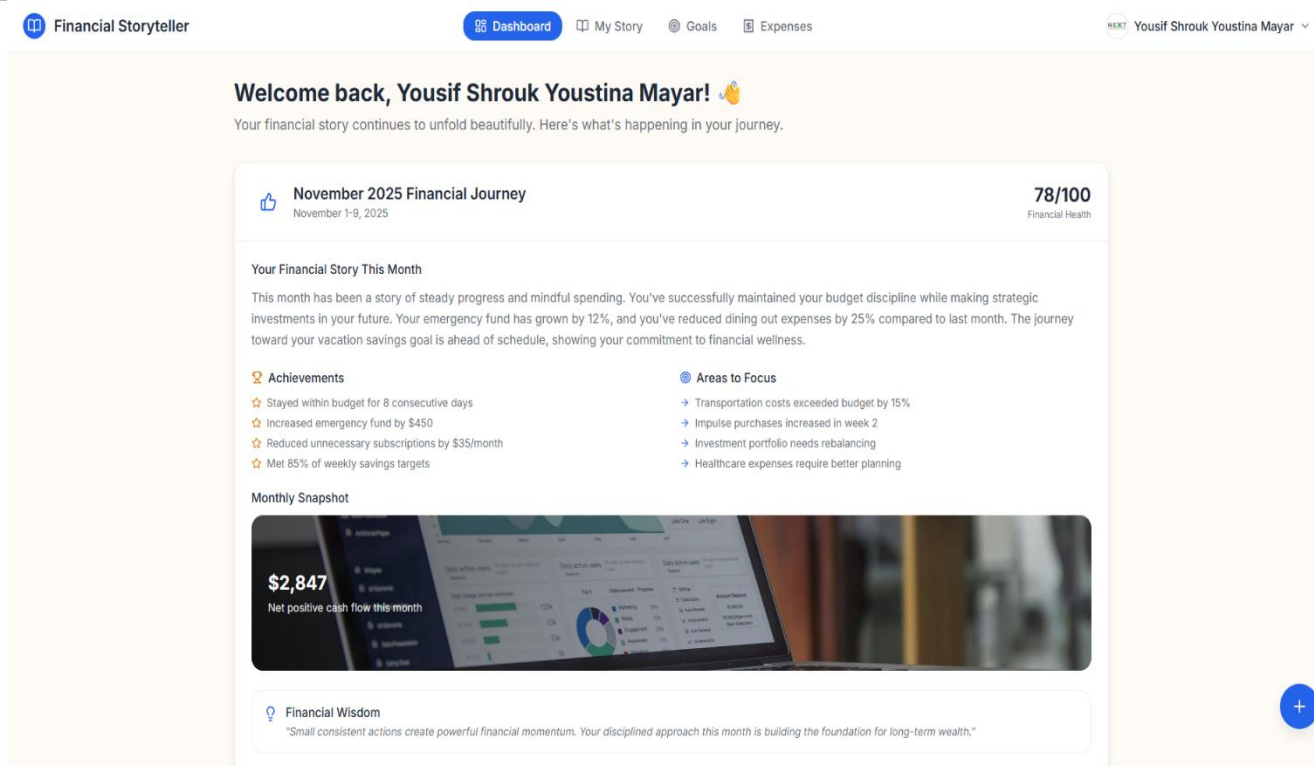


2025

The Financial Storyteller



NXT24_AST3_SWD2_G1 - React
DEPI - Next Academy
11/24/2025

1. Executive Summary

The Financial Storyteller is an innovative personal finance management system designed to transform ordinary financial data into emotionally engaging narratives. Unlike traditional finance applications that present users with spreadsheets, mathematical charts, or overwhelming dashboards, this project introduces a human-centered approach where financial activity becomes a compelling story. Instead of analyzing impersonal numbers, users engage with personalized financial chapters that illustrate their past behavior, present habits, and future trajectory in a form that is emotionally relatable and intellectually enlightening.

This project aims to create a web-based prototype using React.js and JSON storage. It focuses on simplifying financial interpretation for everyday users, increasing engagement, and providing forecasts in a language users understand. Through visually rich timelines, adaptive storytelling, and interactive goal-setting, the Financial Storyteller bridges the gap between finance and human behavior, transforming financial literacy into an enjoyable experience.

2. Project Overview

2.1. Detailed Project Objectives

The primary objective of the Financial Storyteller is to reinterpret financial analytics in a way that resonates emotionally with users and encourages consistent engagement. Traditional finance tools fail not because they lack data, but because they lack context. Users see numbers, but they don't understand the underlying narrative. This project's objective is to give users a narrative-driven context.

The system aims to:

1. Automatically categorize transactions into meaningful financial events.
2. Convert financial data into detailed narrative descriptions using a rule-based narrative engine.

3. Present finances as chapters—past, present, and future—each reflecting a particular phase in the user’s financial life.
 4. Allow users to edit financial goals and immediately see how the narrative changes.
 5. Introduce an emotionally relatable financial learning experience to motivate long-term improvements.
 6. Provide a visually intuitive timeline where stories evolve over weeks, months, or years.
 7. Build a functional prototype that demonstrates the validity of narrative-based personal finance software.
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2.2. Project Vision

The vision behind this application is to humanize personal finance. The system does not simply report information; it interprets it. It acts as a mentor, a storyteller, and a guide that walks the user through their financial life in a tone that inspires self-improvement rather than guilt or anxiety. The ultimate vision is to create a financial tool people actually enjoy using.

2.3. Scope of the Project

The scope includes:

Functional Scope

- Constructing a React.js-based interface structured like a storybook.
- Building a narrative generation engine that reads financial data and transforms it into human-like written stories.
- Designing a timeline component that displays the evolution of the user's financial story.
- Implementing goal-setting functionality where users specify objectives and the system adjusts future narrative predictions accordingly.

- Creating JSON data structures to simulate transactions, user profiles, categories, and story outputs.

Technical Scope

- Data will be stored in structured JSON files.
- No real backend service is required for the prototype; instead, the system uses static JSON assets to simulate database operations.
- The narrative engine will operate using deterministic rules that interpret financial data patterns.

Non-functional Scope

- Ensuring the interface is intuitive and visually readable.
- Offering responsiveness for desktop and tablet.
- Maintaining code that is modular, well-documented, and scalable for future backends.

Out-of-Scope Items (Future Work)

- Real bank API integrations.
- AI-driven natural language generation.
- User authentication, encryption, and authorization.
- Mobile app versions.
- Cloud database storage.

2.4. Success Criteria

1. The success of this project is determined by the system's ability to produce coherent, meaningful narratives that accurately reflect financial activity. The prototype will be considered successful if:
2. The narrative engine produces readable, human-like summaries.
3. Users can clearly navigate through past, present, and future financial chapters.
4. The timeline interface visually communicates financial progression.

5. JSON files are processed correctly, ensuring smooth story generation.
 6. Goal changes produce immediate, consistent updates in the future chapter narrative.
 7. The system demonstrates potential for academic, commercial, or personal use.
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3. Problem Statement

Financial data is naturally abstract, technical, and often overwhelming, especially for users with minimal financial literacy. Many personal finance applications rely heavily on numerical charts, balance sheets, and transaction logs that require interpretation skills most users do not possess. As a result, many people abandon financial management tools due to confusion, anxiety, or lack of motivation.

Moreover, financial data is static—it reflects what happened, not how the user feels or what they should do next. The emotional disconnect between the user and their financial information creates disengagement. Financial problems tend to worsen when people avoid checking their accounts, avoid budgeting, or fail to plan their goals.

There is a clear need for a system that does more than present information: it must interpret it, narrate it, and guide the user in a way that is empathetic and easy to understand. Users require a system that not only informs them but also inspires them.

The Financial Storyteller addresses these issues by translating financial behavior into emotionally meaningful narratives, making finance feel more human, accessible, and engaging.

4. Proposed Solution

The proposed solution is a narrative-driven financial management application designed to interpret numerical financial data and retell it through a storytelling lens. The system makes finance feel more familiar and less intimidating by framing financial events as part of a story. This increases emotional engagement and encourages users to take control of their financial habits.

The solution incorporates:

Narrative Generation Engine

A specialized module that analyzes spending habits, income streams, categories, and patterns to build compelling story chapters. It produces:

- Past financial stories (monthly retrospectives)
- Present status updates (ongoing financial health narrative)
- Future simulation chapters (predicted outcomes based on spending trends and goals)

Interactive Storybook Interface

The frontend includes a visual timeline where each chapter expands into a full narrative experience. The interface resembles a digital book where each month is a chapter, and each financial event is a highlighted paragraph.

Goal-Oriented Financial Planning

Users can create or modify goals. When a goal is added, such as saving a specific amount or reducing a spending category, the future chapter updates to describe whether the user is likely to achieve that goal and how they can improve.

Data Categorization Engine

The system automatically categorizes transactions (e.g., Food, Transport, Utilities). These categories feed into the narrative generator, which transforms raw spending patterns into recognizable story motifs:

- “You indulged more in eating out this month...”
- “Transport expenses remained consistent...”
- “Your savings discipline improved noticeably...”

Predictive Analytics

Using straightforward heuristics or simple formulas, the system can forecast:

- Future expenses
- Goal achievement probability
- Savings acceleration or decline

5. Technical Architecture

The system architecture is designed around separation of concerns, clarity, and future scalability. Each module operates independently but communicates through structured JSON data.

5.1. Frontend Architecture (React.js)

The frontend contains:

- Reusable components (StoryCard, TimelineNode, GoalEditor, AlertsWidget)
- A global state management system (Context API or Redux)
- A narrative rendering engine that dynamically displays story chapters
- Routing for pages such as Home, Timeline, and Goals
- Visual enhancements such as animation, transitions, and expandable story sections

5.2. Narrative Engine Architecture

The narrative engine functions as follows:

1. **Data Retrieval Layer** reads user transactions, categories, and goals from JSON files.
2. **Analysis Layer** processes the raw data to identify:
 - spending trends
 - category dominance
 - financial stability indicators
 - savings habits
3. **Story Synthesis Layer** uses templates and rules to convert analysis results into readable paragraphs.
4. **Tone Modulation Layer** adjusts tone based on user performance (e.g., encouraging, neutral, corrective).
5. **Output Layer** structures the results in an object suitable for rendering by the UI.

5.3. Data Storage Layer

Because this is a prototype, JSON files store:

- User profiles
- Transactions
- Categories
- Goals
- Narrative outputs

Although simple, this structure simulates a real database and allows easy portability to a backend like Node.js, Laravel, or Firebase.

5.4. Integration Logic

The React frontend retrieves JSON data either through:

- Local fetch requests (static hosting)
- Mock API endpoints (if Node.js is used)

The narrative engine then processes this data and pushes the results back into the UI modules.

6. Project Planning

The planning for the Financial Storyteller involves multiple development phases, each with specific deliverables and dependencies. The project follows a semi-agile approach with iterative testing and progressive refinement.

6.1. Development Phases

Phase 1: Research and Requirements

This phase involves studying existing personal finance applications, identifying their weaknesses, and determining how a narrative-based system provides a superior experience. User interviews and market research help shape the SRS (Software Requirements Specification) and define the narrative engine rules.

Phase 2: UI/UX Design

During this phase, the interface layout is conceptualized. Designers create wireframes and high-fidelity mockups that illustrate how the timeline, storycards, narrative chapters, and goals editor will appear. This stage also defines color schemes, typography, spacing, and the storytelling visual identity.

Phase 3: Narrative Engine Architecture

Before implementing UI, the backend logic for converting data into narratives must be designed. This phase includes:

- Designing narrative templates
- Building categorization logic
- Defining narrative tone rules
- Designing story construction pipelines

Phase 4: Frontend Implementation

Developers build the React components and import mock JSON data. The UI framework, routing, and state management are completed. Components begin communicating with the narrative engine.

Phase 5: Optimization and Animations

Animations such as sliding stories, expanding chapters, and smooth goal transitions are implemented to create an engaging experience.

Phase 6: Deployment

The final stage includes:

- Building the React bundle
- Hosting the site on Netlify
- Preparing project documentation

7. Stakeholder Analysis

To ensure the success of The Financial Storyteller, it is essential to clearly identify every individual or group that has a direct or indirect influence on the project. Stakeholders determine requirements, influence priorities, shape constraints, and play a major role in decision-making. Below is a fully expanded, professional, and deeply detailed analysis.

7.1. Stakeholder Identification and Detailed Responsibilities

7.1.1. End Users

End users are the primary beneficiaries of the system. They are individuals who want to track their finances in a more emotionally engaging, narrative-based format rather than using traditional spreadsheet-like dashboards. Their responsibilities include providing financial data (manually or via future integrations), interacting with the story-driven interface, and giving real-time feedback that influences further enhancements. Their needs primarily dictate UI/UX decisions, narrative tone customization, and user experience improvements.

7.1.2. Product Owner

The product owner is responsible for ensuring that the project vision remains aligned with user needs and business goals. This role prioritizes features in the roadmap, validates the usefulness of functionalities, and continuously refines the core value proposition of the application. They communicate directly with stakeholders, especially developers, to clarify ambiguities, approve user stories, and verify that the implemented features match the intended experience.

7.1.3. Project Manager

The project manager oversees the planning, scheduling, and execution of the project. Their responsibilities include managing risks, ensuring the timely delivery of milestones, allocating resources efficiently, and coordinating communication among the development team. They must ensure the project stays within scope and budget while maintaining quality.

7.1.4. UI/UX Designer

The designer ensures that the emotional storytelling nature of the product is reflected in the interface. This includes designing user flows, mapping wireframes, and creating intuitive journey paths where users can navigate through their “financial life chapters.” While this documentation reserves empty sections for your screenshots, the designer’s work essentially shapes how users emotionally experience and interpret their financial data.

7.1.5. Frontend Developer

The frontend developer implements the UI using React.js, adding interactive components such as animated story chapters, timeline navigation, graphs, and dynamic narrative rendering. They work closely with the backend developer to consume APIs and transform raw data into user-friendly storytelling visuals. The frontend developer must ensure performance, accessibility, responsiveness, and smooth animations.

8. Requirements Specification

This section defines exactly what the system should do (functional requirements) and how it should behave in terms of performance, quality, and constraints (non-functional requirements). Each requirement is explained in a deeply detailed manner to avoid ambiguity.

8.1. Functional Requirements

8.1.1. Transaction Input and Storage

The system must allow users to manually enter financial transactions. Each transaction should include an amount, category, date, description, and optional notes. The system must store this data in

a JSON-based data model, ensuring integrity and easy retrieval. Future versions will allow API-based retrieval from bank integrations.

8.1.2. Automatic Transaction Categorization

The application must analyze transaction descriptions and classify them into categories such as Food, Transport, Bills, Entertainment, Savings, or Custom Categories. This categorization uses rule-based logic initially, with future plans for machine learning classification. Categories must influence the narrative tone (e.g., overspending stories, celebration stories for savings, etc.).

8.1.3. Narrative Generation Engine

The core system must translate numerical financial behaviors into coherent storytelling paragraphs. For example:

Instead of: You spent \$300 on dining.

It should generate:

"This month, your dining habits took center stage, with a noticeable rise in café visits and weekend outings. It added color to your month, but you might want to balance it next month."

Narratives must include patterns, warnings, achievements, and predictions about future behavior.

8.1.4. Predictive Analysis

The system must use linear forecasting or rule-based prediction to estimate future spending.

Predictions will be translated into future story chapters: "Next Month's Outlook," "Predicted Spending Behavior," and "Financial Warnings."

8.1.5. Interactive Timeline Interface

Users must be able to navigate through their "Financial Story" using a chapter-like timeline. Each chapter represents past months, current month, or future expectations. Clicking a chapter must dynamically load narratives and visualizations.

8.1.6. Financial Goals and Editing

The system must allow users to set monthly or yearly goals (e.g., savings targets). The narrative engine must adjust the storytelling according to whether the user is on track or behind.

8.1.7. Narrative-Based Notifications

Users must receive small pop-up tales summarizing important insights, e.g., “You just crossed your weekly dining budget — here’s the story so far...”

8.2. Non-Functional Requirements

8.2.1. Performance

The system must load narratives within 300ms on a high-end device and under 1 second on average connections. Graph interactions must remain smooth at 60 FPS.

8.2.2. Scalability

Even though initial storage is file-based, the architecture must support easy migration to MongoDB or PostgreSQL for scalability.

8.2.3. Reliability

Narrative generation must never fail silently. If predictions fail, the system displays a fallback narrative (“We’re preparing your story...”).

8.2.4. Usability

The UI must be accessible, readable, and visually storytelling-oriented, with animations and smooth transitions.

9. UI/UX Design

9.1. Dashboard

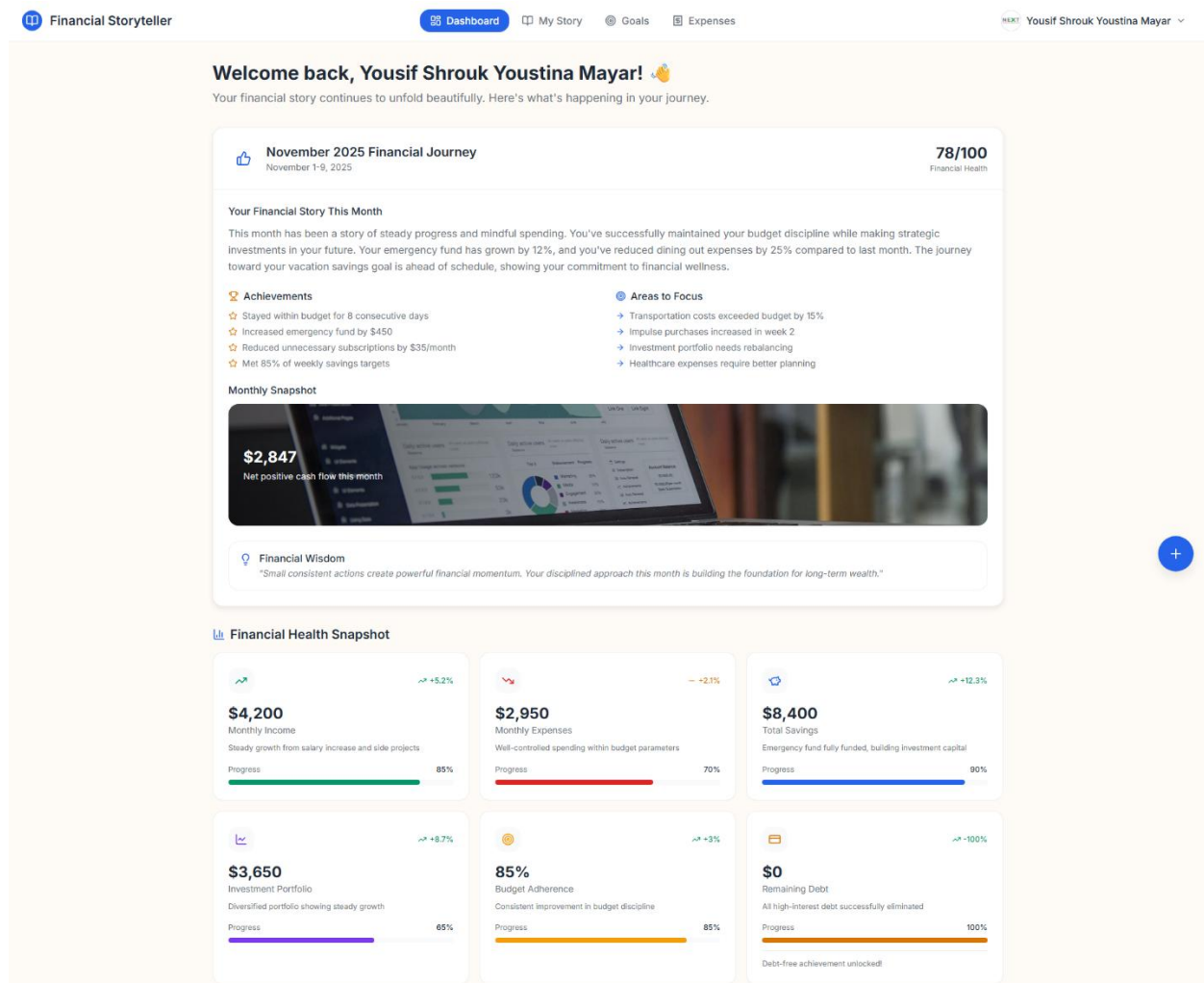


Figure 1: Dashboard

Your Financial Story Chapters


Your Financial Past
January - October 2023

Progress

85%

\$3,200

Debt Paid Off

\$5,800

Savings Built

Story Summary
 Your financial journey began with determination and has evolved into a story of growth and learning. From paying off \$3,200 in credit card debt to building your first emergency fund, you've transformed challenges into stepping stones. Each month brought new lessons about budgeting, saving, and the power of compound interest.

Explore Your Financial Past Story →


Your Current Chapter
November 2025

Progress

72%

\$4,200

Monthly Income

\$2,950

Expenses

Story Summary
 Right now, you're in the momentum phase of your financial story. Your disciplined approach is paying dividends, with consistent savings and smart spending choices. This month represents a turning point where financial wellness becomes a natural part of your lifestyle rather than a struggle.

Explore Your Current Chapter Story →


Your Financial Future
December 2025 - Beyond

Progress

45%

45%

Goal Progress

Dec 2026

Target Date

Story Summary
 The next chapters of your financial story are filled with exciting possibilities. With your strong foundation, you're positioned to achieve major milestones: homeownership, investment growth, and financial independence. Your future self will thank you for the disciplined choices you're making today.

Explore Your Financial Future Story →

Quick Actions

Continue your financial journey

 **Add Expense** →
Record a new expense

 **My Story** →
View financial timeline

 **Set Goals** →
Plan your future

 **Manage Expenses** →
Review & categorize

Recent Activity

	Grocery Shopping	\$85.50
	Gas Station	\$45.00
	Coffee Shop	\$12.75

Financial Alerts

Stay informed about your financial progress

 **Savings Milestone Achieved!** 5m ago ×
Congratulations! You've reached 75% of your emergency fund goal. Your financial security is getting stronger with each contribution.

Context: Emergency fund target: \$10,000 | Current: \$7,500

[View Progress](#) [Adjust Goal](#)

 **Budget Alert: Transportation** 30m ago ×
Your transportation spending is 15% over budget this month. Consider carpooling or using public transit to get back on track.

Context: Budget: \$300 | Spent: \$345 | Remaining: -\$45

[Review Expenses](#) [Adjust Budget](#)

 **Vacation Fund Progress** 1h ago ×
You're ahead of schedule! Your vacation fund has reached \$1,200, putting you 20% ahead of your timeline.

Context: Target: \$2,000 by March 2025

[View Timeline](#)

Progress 60%



Your Financial Journey is Inspiring! ✨

Every dollar saved, every goal achieved, and every wise decision you make is writing a beautiful story of financial success. Keep going - your future self will thank you for the choices you're making today.

Figure 2: Dashboard

9.2. Interactive Timeline

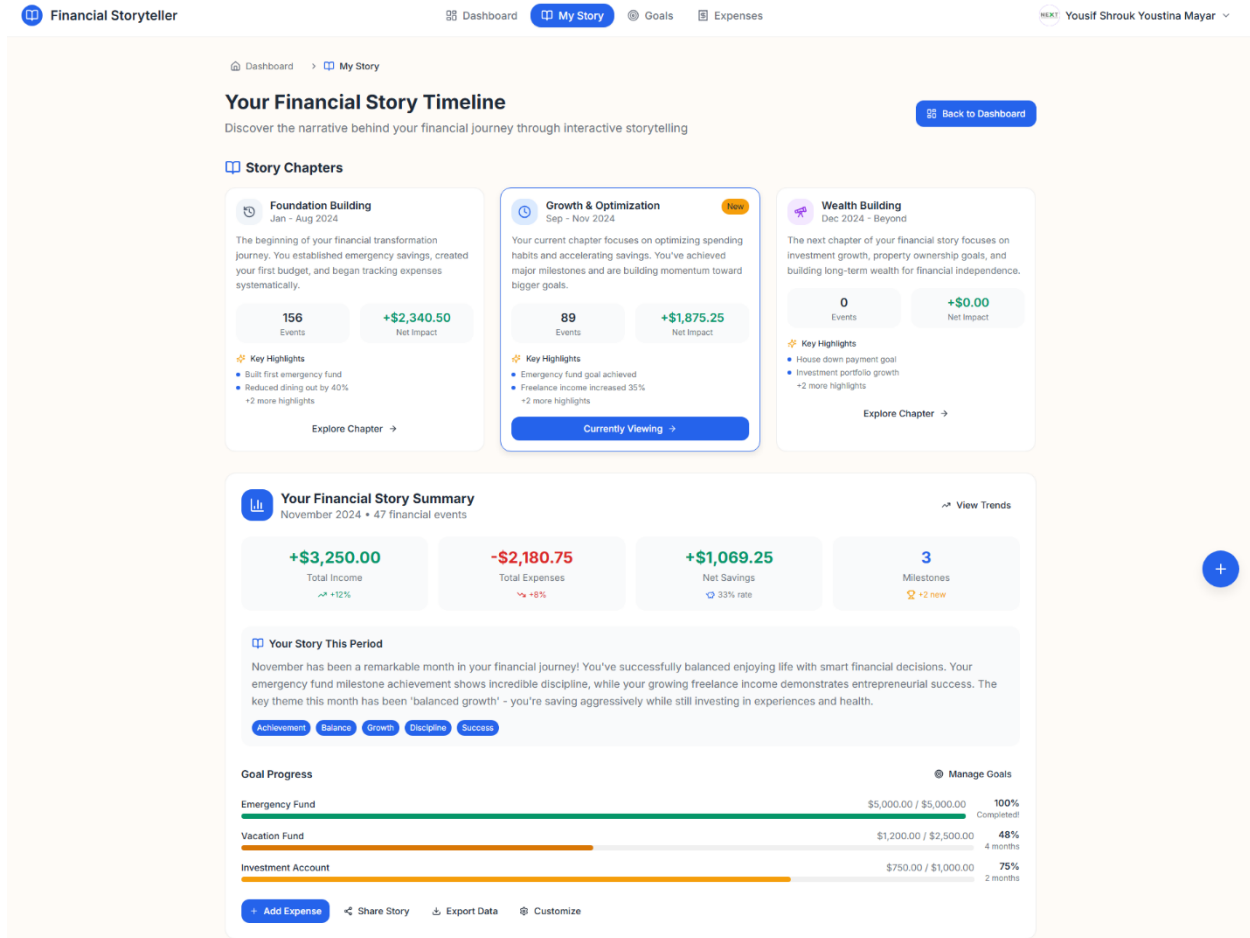


Figure 3: Interactive Timeline

9.3. Narrative Chapter View

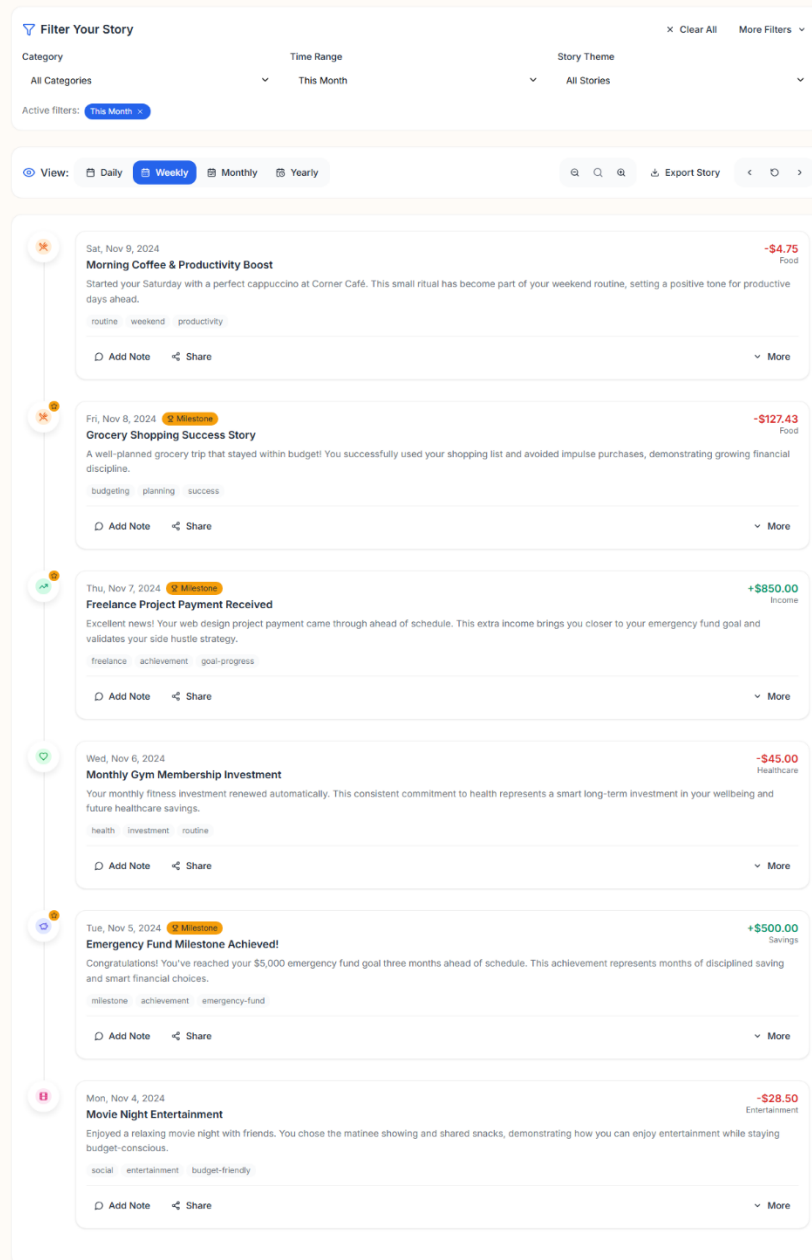


Figure 4: Narrative Chapter View

9.4. Goals and Planning

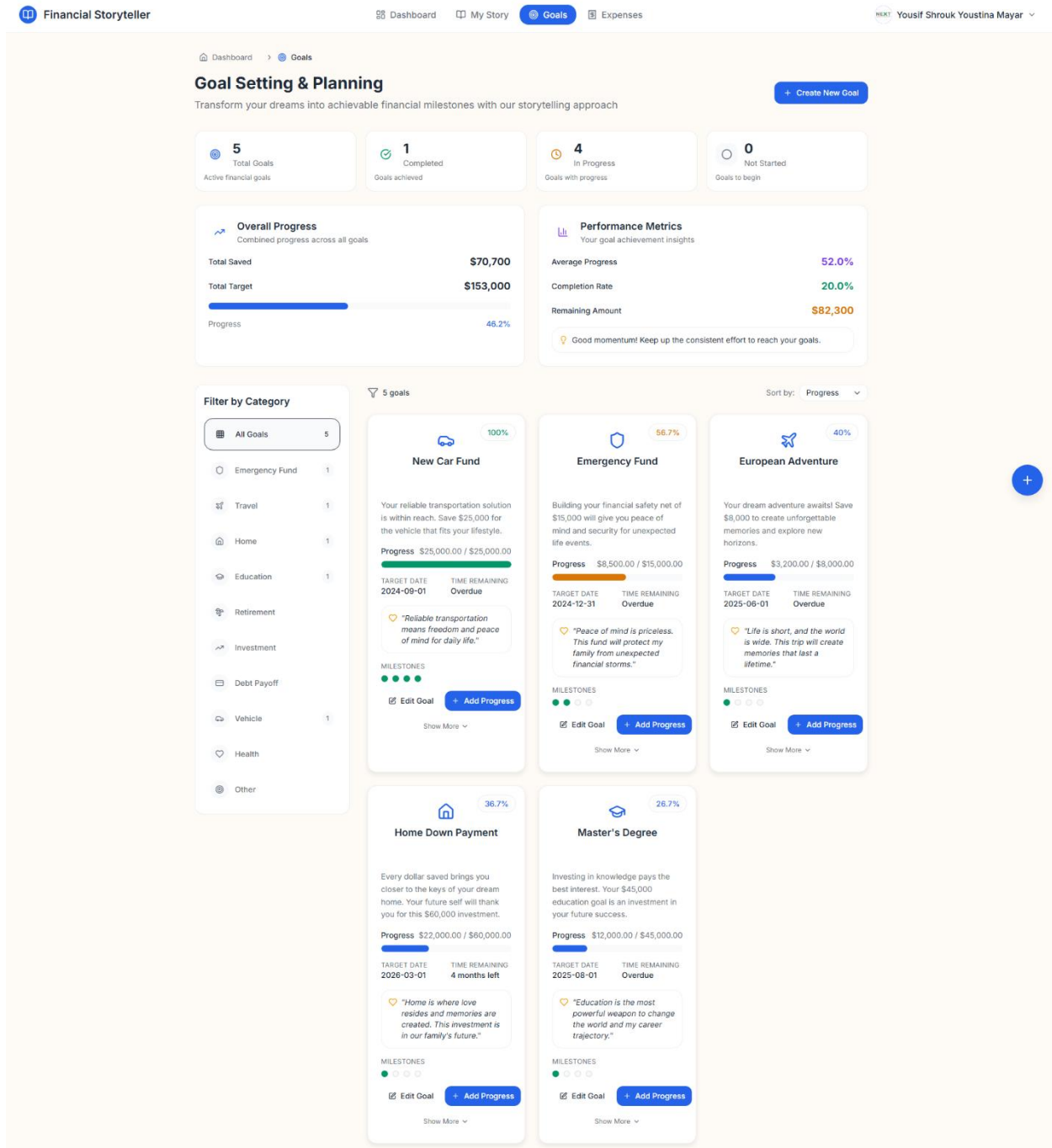
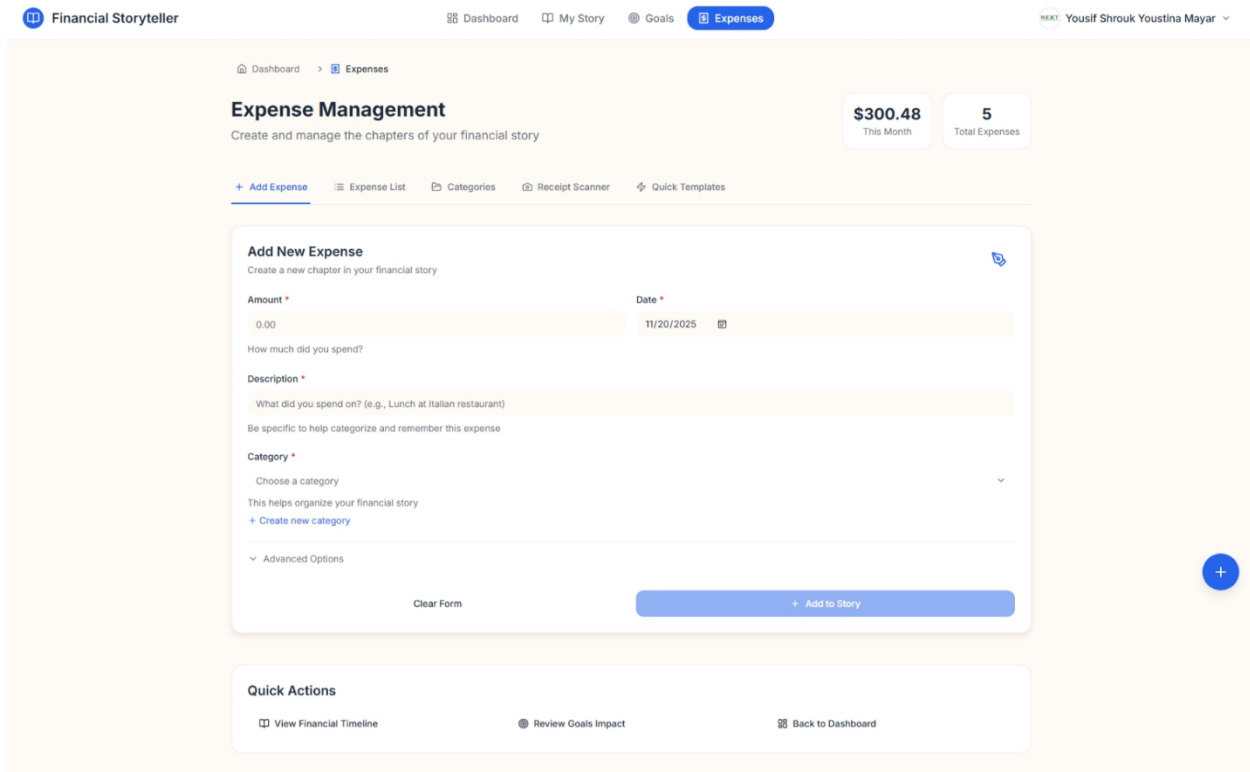


Figure 5: Goals and Planning

9.5. Adding Expenses



Financial Storyteller | Dashboard | My Story | Goals | **Expenses** | Youself Shrouk Youstina Mayor

Dashboard > Expenses

Expense Management

Create and manage the chapters of your financial story

\$300.48 This Month | **5** Total Expenses

+ Add Expense | Expense List | Categories | Receipt Scanner | Quick Templates

Add New Expense

Create a new chapter in your financial story

Amount * 0.00 | Date * 11/20/2025

How much did you spend?

Description *
What did you spend on? (e.g., Lunch at Italian restaurant)
Be specific to help categorize and remember this expense

Category *
Choose a category
This helps organize your financial story
[+ Create new category](#)

Advanced Options

Clear Form | + Add to Story

Quick Actions

View Financial Timeline | Review Goals Impact | Back to Dashboard

Figure 6: Adding Expenses

9.6. Expenses List

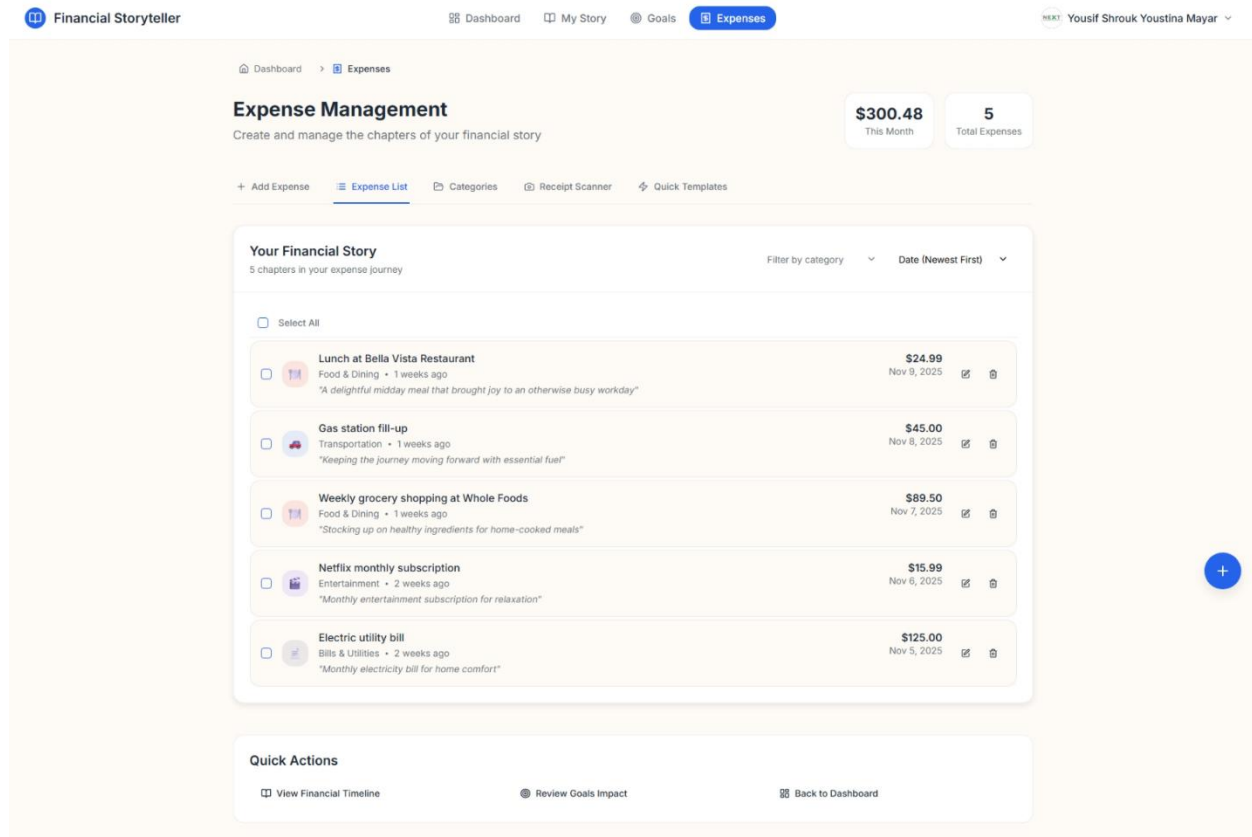


Figure 7: Expenses List

10. Future Enhancements

Potential future features:

- AI-powered narrative personalization,
- ML-based prediction models,
- bank API integrations,
- multilingual support,
- premium subscription tiers,
- integration with smart home financial assistants,
- exporting stories as illustrated eBooks,
- fully gamified financial journeys.